

Plate Reader Set Up

Day 1 – Pre-Culture Setup

- From Petri plate —> inoculation into tubes containing **SC + 2% Glucose**.
- **In the evening:** Inoculate fresh cultures for the next day, aiming for **OD<0.4 by 9:00 AM** on Day 2.

Day 2 – Pre-Growth in Target Medium

- **Morning:** Measure OD. If OD<0.4, continue growing if needed.
- Once cultures reach proper OD (around 0.3), **change medium** to the **target sugar condition** (SC + 2% Glucose, Galactose, or Raffinose) and dilute down to 0.05/0.1.
- This is the **pre-growth condition** (in the scheme this is the first sugar that appears on each well label).
- **Evening:** Inoculate again for the next day, aiming for **OD<0.4 by morning**.
SSSS

Day 3 – Experiment Setup and Start

- **Morning:** Measure OD and dilute cultures.
- **Wash cells** (e.g., centrifuge and resuspend) in **SC without carbon source**.
- Adjust OD to **0.05** in the washed cultures.
- In a microplate, mix:
 - **200 µL of washed culture** with
 - **200 µL of SC medium containing 4% target sugar** (Glucose, Galactose, or Raffinose).
- Final concentration in each well:
 - **OD = 0.025**
 - **Sugar = 2%**
- **Run the experiment** for ~41 hours at **30°C** with **shaking during readings**.

Scheme:

Biological Replicate 1	1	2	3	4	5	6	7	8	Biological Replicate 2
A	X*	X*	X*	GAL GAL	GAL GAL	X*	X*	X*	
B	GLU GAL	GLU GAL	GAL GLU	GAL GLU	GAL GLU	GAL GLU	GLU GAL	GLU GAL	
C	GLU RAF	GLU RAF	RAF GLU	RAF GLU	RAF GLU	RAF GLU	GLU RAF	GLU RAF	
D	RAF GAL	RAF GAL	GAL RAF	GAL RAF	GAL RAF	GAL RAF	RAF GAL	RAF GAL	
E	GLU GLU	GLU GLU	GAL GAL	RAF RAF	RAF RAF	GAL GAL	GLU GLU	GLU GLU	
F	X	X	X	RAF RAF	RAF RAF	X	X	X	

Legenda:

X* = SC with carbon source (SC+4%Glu, SC + 4%Gal, SC + 4%Raf respectively), without cells added.

X = SC no carbon source

In purple = Inductions

In grey = decays

Conditions – dictionary format:

conditions = {

'GLUGLU': ['E1', 'E2', 'E7', 'E8'],

'GALGAL': ['E3', 'A4', 'E6', 'A5'],

'RAFRAF': ['E4', 'F4', 'E5', 'F5'],

'GLUGAL': ['B1', 'B2', 'B7', 'B8'],

'GLURAF': ['C1', 'C2', 'C7', 'C8'],

'RAFGAL': ['D3', 'D4', 'D5', 'D6'],

'GALGLU': ['B3', 'B4', 'B5', 'B6'],

'RAFGLU': ['C3', 'C4', 'C5', 'C6'],

'GALRAF': ['D1', 'D2', 'D7', 'D8']}]